# MC Vertex Reweighting for PPG12:

A Data-Driven Iterative Method on the Truth Vertex

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**Abstract**

The PPG12 efficiency pipeline must reweight MC to match the measured $p$+$p$ luminous region, which is very different between the two crossing angles ($\sigma \simeq 53$ cm at 0 mrad vs $\sigma \simeq 22$ cm at 1.5 mrad, Piwinski narrowed). The production reco-level weight $f(z_r) = D(z_r)/M(z_r)$ closes the reco vertex distribution exactly, but it leaves the *individual* truth-vertex widths of double-interaction MC events at a geometric "half narrow, half wide" compromise ($\sigma \simeq 47$ cm instead of 22 cm), which in turn biases the truth fiducial cut used by the efficiency denominator. The resulting cross-section bias at 1.5 mrad is +1.4 % — comparable to other PPG12 systematics and well within reach of a correctable method. This note presents the nominal fix: an **iterative data-driven reweight** $w(z)$ applied per truth vertex ($w(z_h)$ for single interactions, $w(z_h)\,w(z_{\mathrm{mb}})$ for doubles). The method uses no physics prior for the target shape — only the empirical width of the data reco distribution as the seed Gaussian, and the 99-th percentile of $|z_r^{\mathrm{data}}|$ as the auto-derived closure window. Applied to the photon10 mix, the method converges cleanly at 0 mrad ($\max|D/M-1| < 1$ % in 8 iterations) and reduces $\chi^2$ in the closure window by a factor $\sim$3100 at 1.5 mrad, closing the bulk of the window to the 1–2 % level inside $|z_r| < 30$ cm and reaching $\sim$11 % at the closure-window edge. The code is a standalone Python package in `efficiencytool/truth_vertex_reweight/`; the per-period `reweight.root` outputs are ready to be consumed by `RecoEffCalculator_TTreeReader.C`.

## Contents

**1 Problem and motivation** **2**

**2 Cross-section impact at 1.5 mrad** **3**

**3 Method: data-driven iterative reweight** **3**

**4 Per-period results** **4**

**5 Physics interpretation** **11**

**6 Caveats and next steps** **12**

**7 Summary** **12**

**A Baseline: events with and without a reconstructed vertex** **13**

**B Discarded methods** **14**

**C Double-interaction reco-truth kernel** **15**

**D Files and code** **16**

# 1 Problem and motivation

PPG12 extracts the isolated prompt-photon cross section from jet-dominated $p$+$p$ triggered data. The cross section is proportional to the measured yield divided by the MC-estimated signal efficiency $\epsilon$. Both the numerator (reco clusters in data, passing the selection) and the denominator (MC truth photons in the fiducial window) inherit the MC vertex distribution — the MC prior on the vertex $z$-coordinate enters via (i) cluster kinematics reconstructed relative to the reco vertex $z_r$, (ii) the $|z_r| <$ `vertex_cut` cut applied in data and MC alike, and (iii) the truth fiducial cut $|z_h| <$ `vertex_cut` used for the efficiency denominator. *Any* data/MC mismatch in the vertex $z$ distribution (reco or truth) therefore feeds directly into $\epsilon$ and the extracted cross section.

**Notation.** Throughout this note, $z_r$ denotes the MBD-reconstructed vertex; $z_h \equiv z_{t,1}$ denotes the primary (hard-scatter) truth vertex; and $z_{\rm mb} \equiv z_{t,2}$ denotes the secondary (minimum-bias) truth vertex, present only in double-interaction events. For double-interaction MC, $z_r \approx (z_h + z_{\rm mb})/2$ up to MBD resolution. Sections 1–2 occasionally use the $(z_{t,1}, z_{t,2})$ pair notation when the algebraic decomposition into symmetric/antisymmetric coordinates is needed; elsewhere we use the $(z_h, z_{\rm mb})$ notation exclusively.

**The two periods are very different.** At the 0 mrad crossing angle the beams overlap along their full length and the luminous region is $\sigma_z \simeq 53$ cm. At 1.5 mrad the beams cross at an angle, the beam-beam overlap is geometrically shortened by the Piwinski factor, and the luminous region collapses to $\sigma_z \simeq 22$ cm. The MC samples used in PPG12 were generated with the full $\sim 65$ cm beam profile regardless of period; the reweight has to correct this downward by nearly a factor of three at 1.5 mrad. This is a very large shape correction and any shortcomings of the reweight method are amplified at 1.5 mrad.

**The production method.** The nominal PPG12 reweight (`efficiencytool/RecoEffCalculator_TTreeReade` lines 1331–1372) uses a 1D reco-vertex weight

$$f(z_r) \;=\; \frac{D(z_r)}{M_{\rm reco}(z_r)},$$

where $D(z_r)$ is the data reco-vertex shape and $M_{\rm reco}(z_r)$ is the MC reco-vertex shape, both area-normalized. Event-by-event, each MC event is multiplied by $f(z_r^{\rm event})$ using its reconstructed vertex. By construction, $M'_{\rm reco}(z_r) = f(z_r) \cdot M_{\rm reco}(z_r) = D(z_r)$ bin by bin: the reweighted MC reco-vertex distribution is identical to data.

**Why the production method is not enough.** $f(z_r)$ closes the reco distribution exactly, but it does not close the individual truth-vertex distributions of double-interaction MC events, and it is those truth vertices that enter the efficiency denominator via the truth fiducial cut. The structural reason is a clean algebraic one: decompose the pair $(z_{t,1}, z_{t,2}) = (z_h, z_{\rm mb})$ of truth vertices for a double-interaction event into a symmetric coordinate $\bar{z} \equiv (z_{t,1} + z_{t,2})/2$ and an antisymmetric coordinate $\varepsilon \equiv (z_{t,1} - z_{t,2})/2$. Under the independent Gaussian sampling of the raw MC, $\bar{z}$ and $\varepsilon$ are statistically independent zero-mean Gaussians with $\sigma_{\bar{z}} = \sigma_\varepsilon = \sigma_{\rm MC}/\sqrt{2}$. The reco vertex is $z_r \approx \bar{z}$ (up to MBD resolution), so the production weight $f(z_r)$ depends *only* on $\bar{z}$ and is flat in $\varepsilon$. After reweighting, $\bar{z}$ matches the target Gaussian $\mathcal{N}(0, \sigma^2_{\rm lum}/2)$ by construction but $\varepsilon$ is unchanged. The primary truth vertex $z_h = \bar{z} + \varepsilon$ is therefore a sum of a narrowed and an un-narrowed Gaussian,

$$\sigma^2(z_h) \;=\; \tfrac{1}{2}\,\sigma^2_{\rm lum} \;+\; \tfrac{1}{2}\,\sigma^2_{\rm MC} \;=\; \tfrac{1}{2}(22.3^2 + 62.7^2) \;\simeq\; (47.0\,{\rm cm})^2. \tag{1}$$

$f(z_r)$ cannot touch $\varepsilon$, so it cannot narrow $z_h$ and $z_{\rm mb}$ individually below this "half-narrow, half-wide" compromise. Any principled fix must therefore act on the truth vertices directly, i.e. it must factorize as $w(z_h)\,w(z_{\rm mb})$ rather than depending only on the reco combination $z_r$. The nominal method of Section 3 does exactly this, with a data-driven $w(z)$ iterated to closure on the mixed-MC reco distribution.

## 2 Cross-section impact at 1.5 mrad

The efficiency denominator in `RecoEffCalculator_TTreeReader.C` uses both a reco-vertex cut $|z_r| <$ `vertex_cut` and a truth-vertex cut $|z_h| <$ `vertex_cut`, with `vertex_cut` $= 60$ cm in `config_bdt_nom.yaml`. The reco cut is handled correctly by $f(z_r)$ because the reco distribution is closed to data exactly. The truth cut is *not* handled correctly because, as Eq. (1) shows, $f(z_r)$ leaves $\sigma(z_h) \simeq 47$ cm instead of the target 22 cm for double-interaction MC.

**Quantitative estimate.** For a Gaussian truth distribution of width $\sigma$, the fraction in $|z_h| <$ 60 cm is $\mathrm{erf}(60/(\sigma\sqrt{2}))$. Plugging in:

- $\sigma = 47$ cm (what nominal $f(z_r)$ gives): $\mathrm{erf}(60/(47\sqrt{2})) \simeq 0.80$
- $\sigma = 22$ cm (correct 1.5 mrad): $\mathrm{erf}(60/(22\sqrt{2})) \simeq 0.99$

The double-interaction MC loses $\sim 19$ percentage points of truth fiducial acceptance that it shouldn't. The mixed MC blend at 1.5 mrad is 0.921single+0.079double, so the bias on the integrated efficiency denominator is

$$0.079 \times 0.19 \;\simeq\; 1.4\,\% \quad \text{at } 1.5\,\text{mrad}.$$

This translates directly to a $+1.4\,\%$ bias on the extracted cross section at 1.5 mrad (the efficiency denominator is too small, so the quoted cross section is too large). At 0 mrad the analogous bias is $\sim 0.3\,\%$ because the MC beam profile is already close to the (wider) 0 mrad luminous region. At the percent level the current PPG12 production is biased at 1.5 mrad because $f(z_r)$ does not correct the individual truth vertices $z_h$ and $z_{\mathrm{mb}}$ of double-interaction events. The bias is comparable to other PPG12 systematics and is easily fixable with a truth-level reweight.

## 3 Method: data-driven iterative reweight

The method derives a truth-vertex weight function $w(z)$ — evaluated at $z_h$ for single-interaction MC and at both $z_h$ and $z_{\mathrm{mb}}$ for double-interaction MC — directly from data, with no Gaussian assumption on the luminous-region shape, by iterating the mixed-MC reco distribution to closure against the data reco distribution.

**Forward model.** Let $f_d$ be the cluster-weighted double-interaction fraction in a given crossing angle (0.079 at 1.5 mrad, 0.224 at 0 mrad) and $f_s = 1 - f_d$ the single-interaction fraction. The mixed MC reco distribution under a candidate truth-vertex weight $w(z)$ is

$$M(z_r) \;=\; f_s \sum_{e \in T_s} w(z_h^e)\, \delta(z_r - z_r^e) \;+\; f_d \sum_{e \in T_d} w(z_h^e)\, w(z_{\mathrm{mb}}^e)\, \delta(z_r - z_r^e), \tag{2}$$

evaluated event-by-event over the single-interaction sample $T_s$ and the double-interaction sample $T_d$. For double-interaction events the weight enters *twice*, once for the primary truth vertex $z_h$ and once for the secondary truth vertex $z_{\mathrm{mb}}$, reflecting the statistical independence of the two truth vertices under the generator. This is the crucial structural choice: it gives the iteration access to the antisymmetric coordinate $\varepsilon$ of Section 1 that the production reco-only weight $f(z_r)$ cannot touch.

**Binning and closure window.** Both periods use the same binning: 80 bins over $[-200, 200]$ cm (bin width 5 cm), no overflow clipping, and identical bin edges for the $w(z)$ weight function and the reco-vertex histograms. The iterative closure window is auto-derived per period as the 99-th percentile of $|z_r^{\mathrm{data}}|$:

- 1.5 mrad: $z_{\mathrm{cut}} = 83$ cm (34 bins inside), from $|z_r^{\mathrm{data}}|_{99\%} = 82.0$ cm.
- 0 mrad: $z_{\mathrm{cut}} = 144$ cm (58 bins inside), from $|z_r^{\mathrm{data}}|_{99\%} = 143.9$ cm.

The window is purely data-driven and requires no manual tuning. Outside it, the statistics in data drop below the level where the iteration could meaningfully constrain $w(z)$.

**Seed.** The iteration starts from a Gaussian seed $w_0(z) = \mathcal{N}(0, \sigma^2_{\text{seed}}) \,/\, \mathcal{N}(0, \sigma^2_{\text{gen}})$, with $\sigma_{\text{gen}}$ measured from the photon10 primary truth vertex $z_h$ (`vertexz_truth` branch) ($\sigma_{\text{gen}} = 64.997 \pm 0.015$ cm, intrinsic to the generator and identical between periods, since the same single MC is loaded for both) and $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ measured directly from the data reco-vertex distribution in the relevant run range (21.48 cm at 1.5 mrad, 54.32 cm at 0 mrad). Note that $\sigma_{\text{seed}}$ requires *no* physics input — no Piwinski narrowing factor, no luminous-region calculation. It is just the empirical width of the reco distribution we are trying to match.

**Iterative update.** At iteration $i$, the residual is $r_i(z_r) = D(z_r)/M_i(z_r)$, where $D(z_r)$ is the data reco-vertex histogram and $M_i(z_r)$ is the mixed MC reco-vertex histogram from Eq. (2) under the current $w_i$. The update is Lucy–Richardson-like, multiplicative, and damped:

$$w_{i+1}(z) \;=\; w_i(z) \,\cdot\, \big[\text{smooth}(r_i(z))\big]^{\alpha}, \qquad \alpha = 0.5, \tag{3}$$

where the residual $r_i$ evaluated at reco bin $z_r$ is applied to the weight at the matching truth bin $z = z_r$ (using the shared binning). After each update, $w$ is renormalized so that $\langle w(z_h) \rangle$ averaged over the single-MC truth distribution equals unity, preserving the physics double-interaction fraction. Damping ($\alpha = 0.5$) prevents over-correction; a 1-bin Gaussian smoothing on the residual prevents bin-by-bin oscillation; the update factor is clipped to $[0.5, 2]$ per iteration to bound jumps; and the endpoints of the update region are edge-clamped to their nearest interior values to prevent runaway behavior at the boundary. The update is applied over the full 80-bin binning and the final $w(z)$ is data-driven everywhere, with no post-iteration snap to a seed Gaussian. (An earlier version of the method snapped $w(z)$ to the seed Gaussian outside the closure window; a systematic study with `compare_snap_strategies.py` showed that removing the snap improves 0 mrad closure by a factor $\sim 1.3$ in $\chi^2$ and $\sim 3$ in $\max|r-1|$ while leaving 1.5 mrad essentially unchanged.) Convergence is tested as $\max|r_i(z) - 1| < 0.01$ inside the closure window, with a hard cap of 30 iterations.

**Implementation.** The method is a standalone Python tool in `efficiencytool/truth_vertex_reweight/`. The mixed-MC histogram is heavy to rebuild ($\sim 22$ M events at 1.5 mrad, $\sim 48$ M at 0 mrad), so per-sample arrays are cached to `output/cache/*.npz` keyed on file mtime; each iteration runs in seconds per iteration on a single core. A short file list is in Appendix D. Per-period outputs live in `output/{1p5mrad,0mrad}/` and include `reweight.root` (histograms `h_w_iterative`, `h_w_seed`, `h_M_iterative`) and `fit_result.yaml`.

# 4 Per-period results

Both periods are fit on the photon10 / photon10_double mix only at this stage; the full 4-sample blend including jet12/jet12_double is a follow-up (Section 6). Table 1 collects the headline numbers from both fits.

| quantity | 1.5 mrad | 0 mrad |
|---|---|---|
| run range | 51274–54000 | 47289–51274 |
| luminosity [$\mathrm{pb}^{-1}$] | 16.8588 | 32.6574 |
| $f_{\mathrm{double}}$ (cluster-weighted) | 0.079 | 0.224 |
| data $\lvert z_r \rvert$ 99% percentile [cm] | 82.0 | 143.9 |
| closure window $z_{\mathrm{cut}}$ [cm] | 83 | 144 |
| bins inside closure window | 34 | 58 |
| $\sigma_{\mathrm{seed}} = \mathrm{std}(z_r^{\mathrm{data}})$ [cm] | 21.48 | 54.32 |
| $\sigma_{\mathrm{gen}}$ from MC $z_h$ (generator) [cm] | $64.997 \pm 0.015$ | |
| initial $\chi^2$ (in window) | 1,419,454 | 26,589 |
| final $\chi^2$ (in window) | 453 | 158 |
| $\chi^2$ reduction factor | 3132× | 168× |
| final $\chi^2$/ndf (in window) | 13.7 (ndf 33) | 2.78 (ndf 57) |
| initial $\max \lvert r-1 \rvert$ | 0.635 | 0.079 |
| final $\max \lvert r-1 \rvert$ (in window) | 0.111 | 0.008 |
| iterations used | 30 (cap hit) | 8 |
| converged to 0.01 tolerance | **no** | **yes** |
| (context) parametric Gaussian $\chi^2$/ndf | 3237 | 454 |

Table 1: Iterative data-driven reweight, photon10 mix only, both crossing angles, with the post-iteration snap to seed removed (Section 3). The 0 mrad fit converges to the 1 % per-bin tolerance in 8 iterations (final $\max \lvert r-1 \rvert \simeq 0.8\,\%$). The 1.5 mrad fit reduces $\chi^2$ in the closure window by a factor $\sim 3100$ in 30 iterations and stalls at $\max \lvert r-1 \rvert \simeq 11\,\%$, dominated by bins at the closure-window edge $\lvert z_r \rvert \simeq 83$ cm. Inside the bulk $\lvert z_r \rvert < 30$ cm the per-bin closure is at the 1–2 % level (see Figure 4). The last row reports the parametric Gaussian $\chi^2$/ndf (best-fit $\sigma_{\mathrm{tgt}}$ from Stage-1) against the data; the iterative method improves on this by a factor of $\sim 236$ at 1.5 mrad and $\sim 163$ at 0 mrad.

**Hero summary.** Figure 1 is the single-figure summary: data reco vs reweighted MC reco overlaid for both periods, residuals, the recovered $w(z)$, and the headline numbers. The 0 mrad mix-MC matches data to $\lesssim 1\,\%$ per bin throughout the closure window; the 1.5 mrad mix matches to $\sim 1$–$2\,\%$ in the bulk $\lvert z_r \rvert < 30$ cm, degrades to $\sim 6$–$8\,\%$ around $\lvert z_r \rvert \simeq 60$ cm, and reaches $\sim 11\,\%$ at the closure-window edge $\lvert z_r \rvert \simeq 83$ cm.

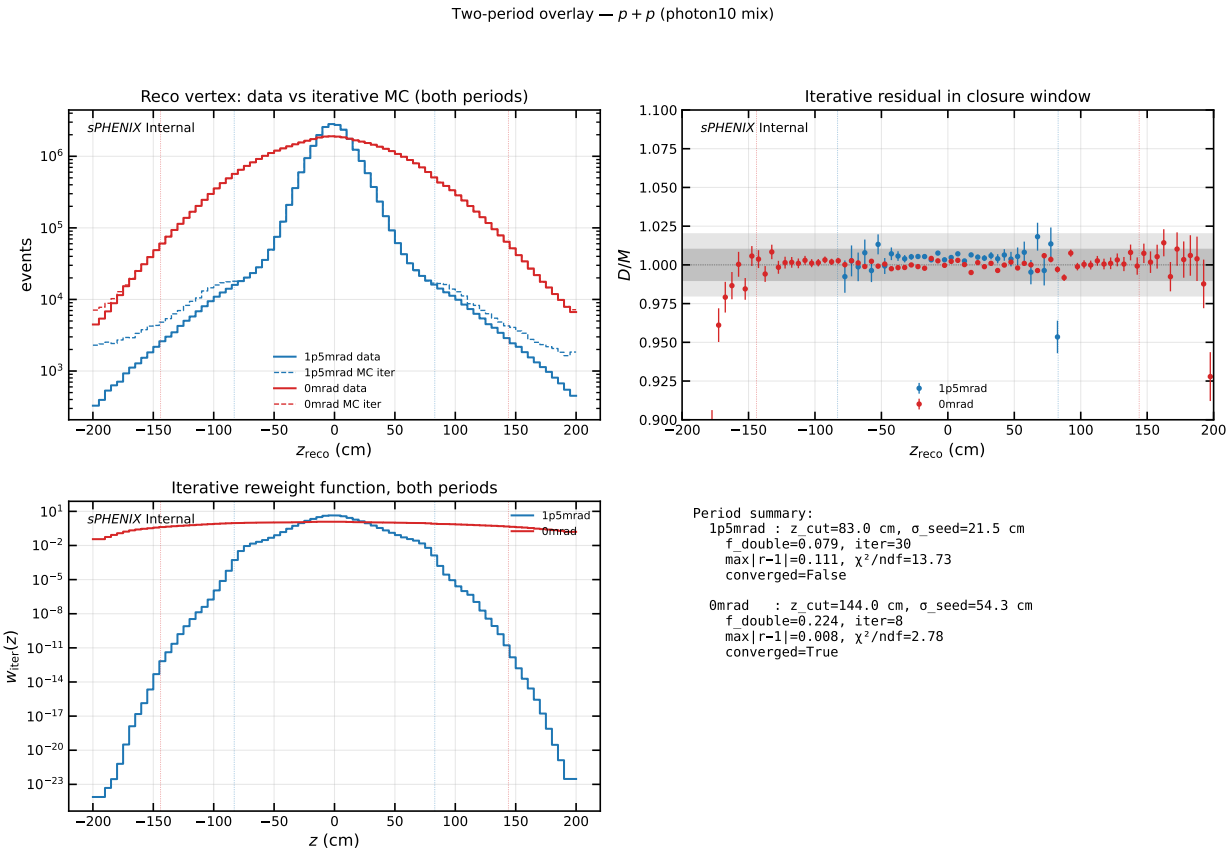


Figure 1: Iterative data-driven reweight summary, both crossing angles ($2 \times 2$ grid plus summary text). Top-left: data reco $z_r$ (solid) vs mixed MC reco $z_r$ after iterative reweight (dashed), both periods overlaid (log scale). Top-right: residual $r(z_r) = D(z_r)/M_{\text{iter}}(z_r)$ inside each period's closure window (vertical dashed lines). Bottom-left: recovered truth-vertex weight $w(z)$ for both periods (log scale). Bottom-right: summary statistics. The 0 mrad case is essentially closed; the 1.5 mrad case is closed in the bulk and stalls at the closure-window edge (see Section 5).

**Truth vertex distribution comparison.** Figure 2 overlays the primary truth vertex $z_h$ distribution for the mixed MC under three weighting methods: no reweight (raw MC, $\sigma \simeq 64$ cm), the production reco-level weight $f(z_r)$, and the new iterative truth-level weight $w(z_h)$. At 1.5 mrad the production method leaves the mixed-MC primary truth vertex $z_h$ at $\sigma \simeq 27$ cm — narrower than the raw MC but still far wider than the true luminous-region width ($\sigma_{\text{lum}} \simeq 22$ cm), because $f(z_r)$ cannot narrow $z_h$ and $z_{\text{mb}}$ individually for double-interaction events below the half-narrow/half-wide compromise (Section 1). The iterative method $w(z_h)$ narrows the $z_h$ distribution to $\sigma \simeq 16$ cm, which is narrower than the data reco width $\sigma_{\text{seed}} = 21.5$ cm precisely as expected, since each single-interaction truth vertex must be narrower than the observed reco width (the MBD vertex finder adds resolution smearing). At 0 mrad, where the correction is small, both methods converge to the same $\sigma \simeq 54$–$57$ cm.

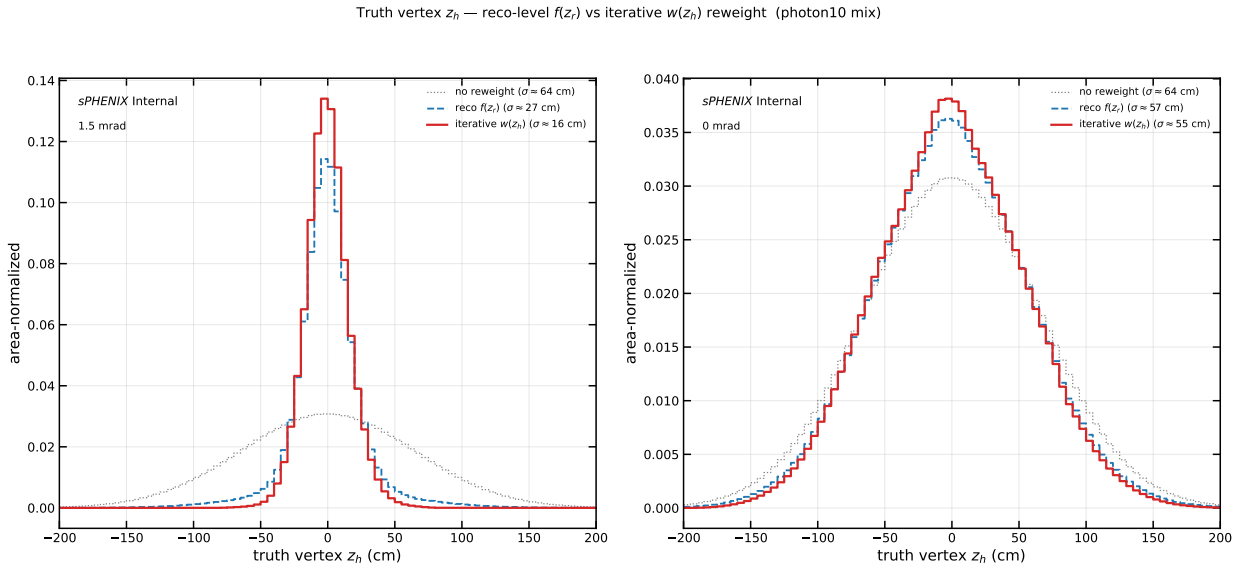


Figure 2: Truth vertex $z_h$ distribution for the mixed photon10 MC under three reweighting methods, per period. Left: 1.5 mrad, where the reco-level $f(z_r)$ (blue dashed) fails to narrow the truth distribution sufficiently, while the iterative $w(z_h)$ (red) narrows it well past the data reco width (the truth must be narrower than the reco due to MBD resolution smearing). Right: 0 mrad, where both methods agree because the MC beam profile is already close to the data luminous region.

**0 mrad: clean convergence.** Figure 3 shows the 6-panel diagnostic for 0 mrad. The seed Gaussian with $\sigma_{\text{seed}} \simeq 54$ cm already captures most of the shape (initial $\max|r-1| = 8\,\%$, initial $\chi^2/\text{ndf} \simeq 467$); the parametric best-fit Gaussian at $\sigma_{\text{tgt}} \simeq 54.6$ cm is essentially the same ($\chi^2/\text{ndf} \simeq 454$) because the data is nearly Gaussian. The iteration cleans up the residual over 8 updates and drops below the 1 % per-bin tolerance inside the closure window (final $\max|r-1| \simeq 0.8\,\%$, $\chi^2/\text{ndf} \simeq 2.78$). The recovered $w(z)$ is smooth and broadly Gaussian, with a small enhancement of the central core relative to the seed, consistent with the data luminous region being slightly more peaked than a single Gaussian.

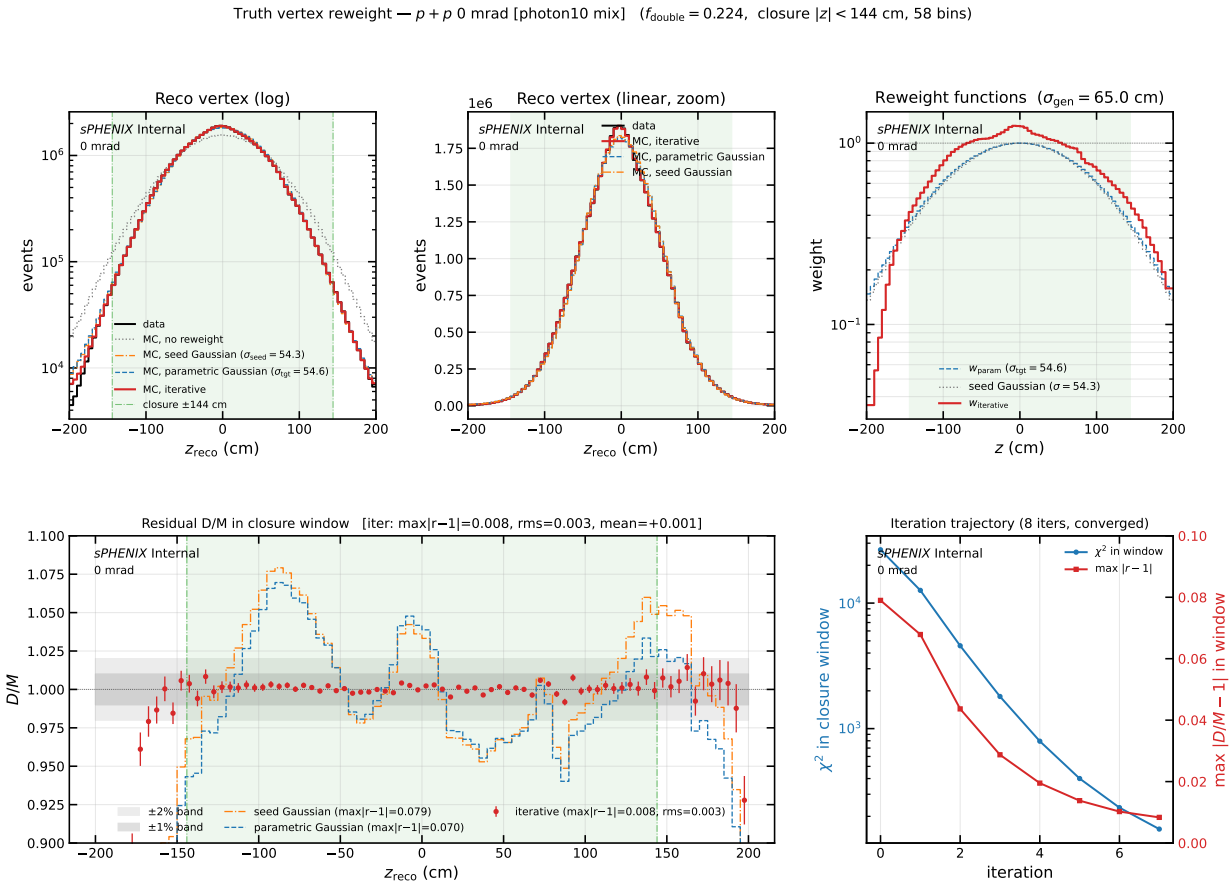


Figure 3: 0 mrad iterative closure (6 panels). Top-left/centre: data reco vs mixed MC reco under no reweight (gray dotted), the seed Gaussian $w(z) \propto \mathcal{N}(0, \sigma^2_{\mathrm{seed}})/\mathcal{N}(0, \sigma^2_{\mathrm{gen}})$ with $\sigma_{\mathrm{seed}} = \mathrm{std}(z_r^{\mathrm{data}})$ (orange dash-dot), the parametric best-fit Gaussian with $\sigma_{\mathrm{tgt}}$ from the $\chi^2$ scan (blue dashed), and the iterative reweight (red). Top-right: the corresponding reweight functions $w(z)$ (seed, parametric, iterative). Bottom-left: residual $D/M$ in the closure window with Poisson error bars on the iterative points and $\pm 1\,\%/\pm 2\,\%$ reference bands. Bottom-right: per-iteration $\chi^2$ and $\max|r-1|$ trajectory. At 0 mrad the seed and parametric Gaussians are nearly identical ($\sigma_{\mathrm{seed}} \simeq \sigma_{\mathrm{tgt}} \simeq 54$ cm) because the data reco distribution is already close to a single Gaussian; the iterative method improves on both.

**1.5 mrad: bulk closure, edge stall.** Figure 4 shows the 6-panel diagnostic for 1.5 mrad. The story is qualitatively different. The seed Gaussian with $\sigma_{\mathrm{seed}} \simeq 21$ cm starts from $\max|r-1| = 64\,\%$ and an in-window $\chi^2$ of $1.42 \times 10^6$; the first $\sim 5$ iterations make dramatic progress, then the iteration enters a slow plateau driven by bins at the closure-window edge $|z_r| \simeq 83$ cm. After 30 iterations, $\chi^2$ in the window has dropped by $\sim 3100\times$ to $\simeq 453$, and $\max|r-1|$ sits at $\simeq 0.11$, set by those stuck edge bins. Inside the bulk $|z_r| < 30$ cm closure is at the $\sim 1$–$2\,\%$ level on every bin; between 30 and 60 cm the residual climbs to $\sim 5$–$10\,\%$. The recovered $w(z)$ shows a strong central core boost and a clear non-Gaussian shape in the $|z| \simeq 40$–$60$ cm region (truth-vertex axis) that no symmetric Gaussian ansatz can represent.

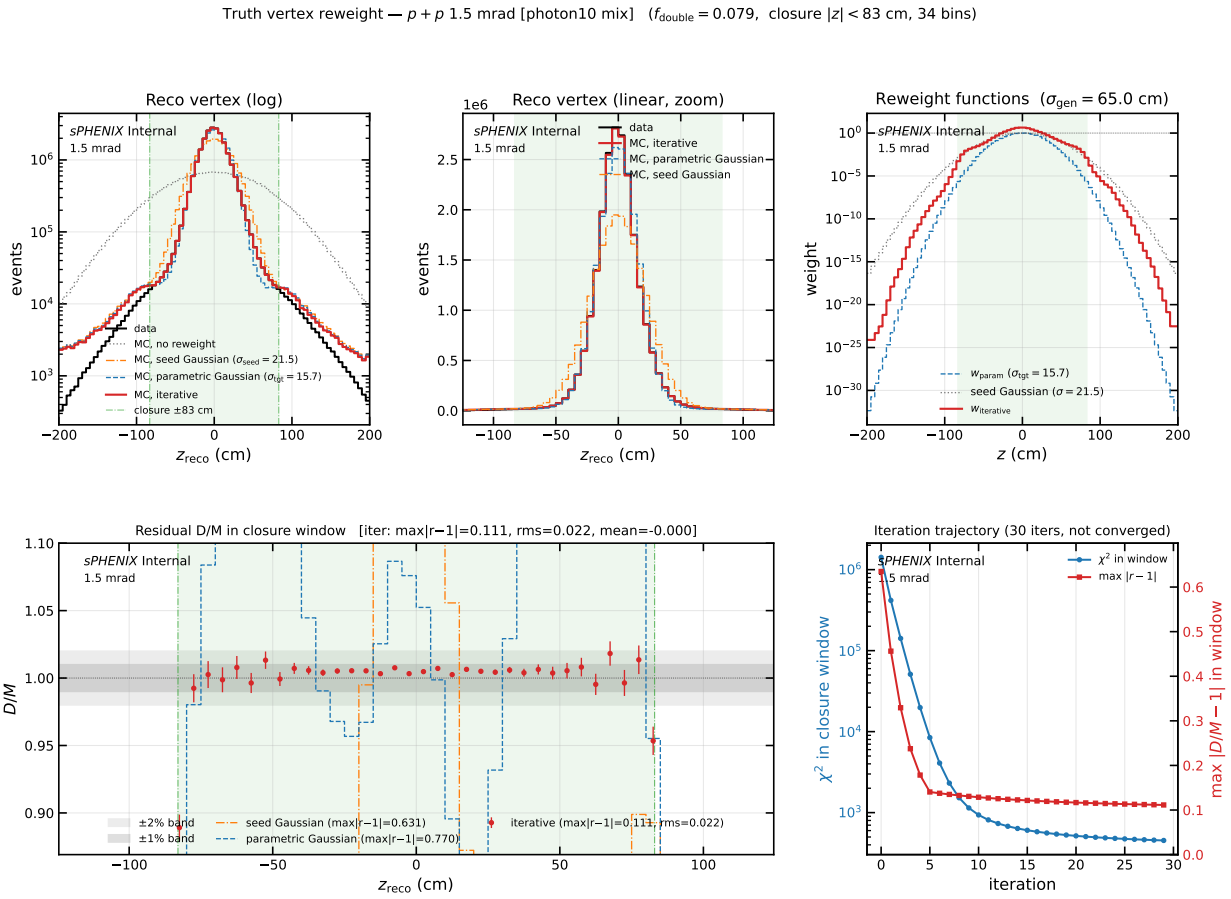


Figure 4: 1.5 mrad iterative closure (6 panels), same layout as Figure 3. At 1.5 mrad the seed Gaussian ($\sigma_{\mathrm{seed}} = 21.5$ cm, orange dash-dot) and the parametric best-fit Gaussian ($\sigma_{\mathrm{tgt}} = 15.7$ cm, blue dashed) are distinct; both are much broader than the data core and mis-predict the reco-vertex shape by $\sim$30–60 % at the closure-window edge. The iterative method (red points) closes the bulk of the window to $\sim$1–2 % inside $|z_r| < 30$ cm and stalls at $\max|r-1| \simeq 0.11$ at the closure-window edge $|z_r| \simeq 83$ cm. The recovered $w(z)$ in the top-right panel shows a strong central core boost and a clear non-Gaussian shape in the $|z| \simeq 40$–60 cm region (truth-vertex axis).

**Method comparison.** Figure 5 puts the iterative reweight against three reference baselines at 1.5 mrad: no reweight, the seed Gaussian (the $\mathcal{N}(0, \sigma^2_{\mathrm{seed}})$ initial point of the iteration, with $\sigma_{\mathrm{seed}} = 21.5$ cm measured from $\mathrm{std}(z_r^{\mathrm{data}})$), and the parametric best-fit Gaussian (a single-parameter fit scanning $\sigma_{\mathrm{tgt}}$ against data, best at $\sigma_{\mathrm{tgt}} = 15.7$ cm). (Stage 1+2, a parametric residual correction on top of the parametric Gaussian, is never used in the nominal pipeline and is not shown.) The no-reweight MC overshoots data by a factor of several everywhere; both Gaussian baselines close the width on average but mismatch the shape at the 30–70 % level, with visible structure in the ratio panel that no single-Gaussian ansatz can represent; the iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

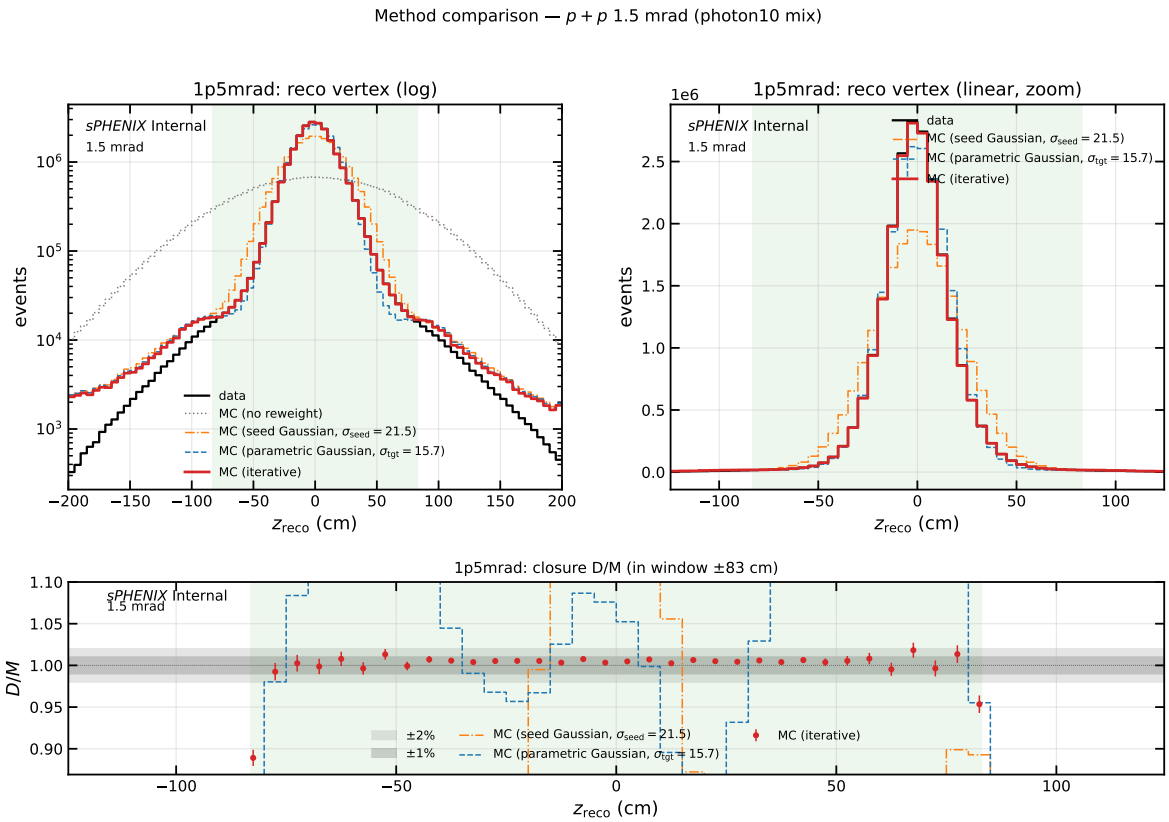


Figure 5: Method comparison at 1.5 mrad: data reco vs mixed MC reco under no reweight (gray dotted), seed Gaussian $\sigma_{\text{seed}} = 21.5$ cm (orange dash-dot), parametric best-fit Gaussian $\sigma_{\text{tgt}} = 15.7$ cm (blue dashed), and the iterative data-driven reweight (red), with a ratio panel below. The two Gaussian baselines bracket the "best possible" single-Gaussian description of the data reco shape; neither closes below the 10–30 % level because the true data-MC ratio at 1.5 mrad is non-Gaussian. The iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

**Parametric $\sigma_{\text{tgt}}$ scan.** Figure 6 shows the stage 1 parametric $\chi^2$ scan that precedes the iterative method. A single Gaussian truth-vertex weight $w(z) \propto \mathcal{N}(0, \sigma^2_{\text{tgt}})/\mathcal{N}(0, \sigma^2_{\text{gen}})$ is evaluated at each scan point. At 1.5 mrad the minimum is at $\sigma_{\text{tgt}} \simeq 15.7$ cm with $\chi^2/\text{ndf} \simeq 3237$; the iterative method reaches 13.7, a $\sim 236\times$ improvement. At 0 mrad the minimum is at $\sigma_{\text{tgt}} \simeq 54.6$ cm with $\chi^2/\text{ndf} \simeq 454$; the iterative method reaches 2.78, a $\sim 163\times$ improvement. The gap between the best parametric point and the iterative floor quantifies the shape mismatch that no single-Gaussian ansatz can capture.

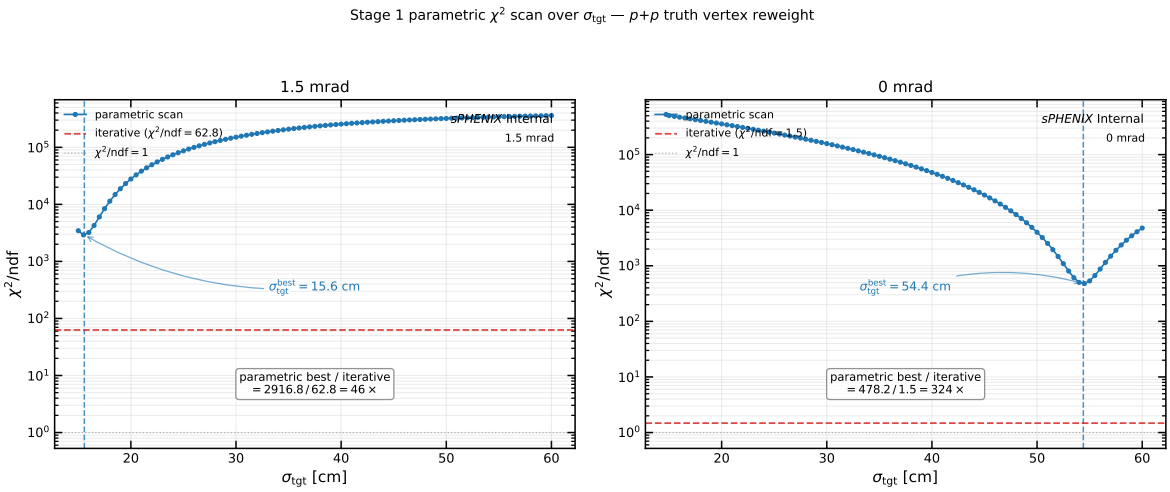


Figure 6: Stage 1 parametric $\chi^2/\mathrm{ndf}$ as a function of the target truth-vertex Gaussian width $\sigma_{\mathrm{tgt}}$, for the 1.5 mrad (left) and 0 mrad (right) periods. The minimum of the scan identifies the best single-Gaussian approximation to the data reco-vertex shape; the parabolic-refined best-fit $\sigma_{\mathrm{tgt}}$ is marked by a vertical dashed line. The horizontal dashed line (red) shows the final $\chi^2/\mathrm{ndf}$ achieved by the iterative data-driven method, and the dotted line marks $\chi^2/\mathrm{ndf} = 1$ for reference. All $\chi^2$ values are computed on the reco-vertex axis $z_r$ inside the closure window and normalized by the number of degrees of freedom.

## 5 Physics interpretation

**Why 0 mrad is easy.** At 0 mrad the beams overlap head-on along their full length and the luminous region is broad ($\sigma_{\mathrm{lum}} \simeq 54$ cm), nearly equal to the generator beam profile ($\sigma_{\mathrm{gen}} \simeq 65$ cm). The shape of the data reco vertex is also close to a single Gaussian. The seed is therefore already a good first approximation, the initial residual is $\sim 8\,\%$, and the iteration just cleans up small departures (slightly enhanced central core). Convergence in a handful of iterations is what one would expect from a near-textbook deconvolution problem.

**Why 1.5 mrad is hard.** Three things conspire at 1.5 mrad:

- The Piwinski crossing-angle effect compresses the luminous region to $\sigma_{\mathrm{lum}} \simeq 21$ cm, almost a factor of three narrower than the generator profile. The reweight must extract a narrow target from a wide source, and the dynamic range of $w(z)$ at the central core is correspondingly large ($w(0) \simeq 8$).
- The data reco distribution is *not* a clean Gaussian. It has a narrow core plus a mid-tail shoulder near $|z_r| \simeq 45$–$60$ cm. The iterative method discovers this structure directly — the recovered $w(z)$ is non-Gaussian in the same $z$ range — whereas any symmetric one-parameter Gaussian ansatz cannot represent it. This is why the parametric best-fit Gaussian gives $\chi^2/\mathrm{ndf} \simeq 3237$ on the 1.5 mrad mix, and why the iterative method drops it to $\simeq 14$ (a further $\sim 236\times$ improvement).
- The MBD resolution tail leak from core-truth events fills the bins near the closure-window edge ($|z_r| \simeq 83$ cm). The iteration becomes insensitive to $w(z)$ updates at that $z$ because the contributions to those reco bins come dominantly from truth vertices ($z_h$, $z_{\mathrm{mb}}$) at other $z$ values via the wide double-interaction reco-truth kernel. The iteration has the wrong handle on those bins, and that is what sets the residual floor at $\max|r-1| \simeq 11\,\%$.

**What the recovered $w(z)$ looks like.** At 1.5 mrad, $w(z)$ has three distinguishing features visible in the top-right panel of Figure 4: (a) a strong central peak with $w(0) \simeq 8$ that pulls double-

interaction events whose truth vertices ($z_h$ and $z_{\mathrm{mb}}$) both lie close to the IP into the data-like core; (b) a mid-$z$ dip that suppresses the generator's natural $|z| \sim 30$–$50\,\mathrm{cm}$ population to match the narrow data core; and (c) a clear $\pm$ asymmetry in $|z| \simeq 40$–$60\,\mathrm{cm}$ tracking the asymmetric mid-tail bump in the data. None of these features is representable by a symmetric single-parameter Gaussian. At 0 mrad, $w(z)$ is essentially the seed Gaussian with a small central enhancement.

## 6 Caveats and next steps

**(i) photon10-only mix.** The current fit uses `photon10 + photon10_double` only. The full production blend also includes `jet12 + jet12_double`. Extending the fit to all four samples is the obvious next step: mechanically it is two more `load_sample` calls plus the appropriate cluster-weighted blend, but the closure numbers will move and should be remeasured before any production hookup.

**(ii) 1.5 mrad is not formally converged.** The 30-iteration cap was hit. Two questions remain open: (a) whether extending to $\sim$100 iterations would push the stuck edge bin to convergence, or whether it is a hard physical floor set by the double-interaction reco-truth kernel (the suspicion is the latter); and (b) whether the failure to converge in $\max|r-1|$ matters for the downstream analysis. The latter has a fairly clean answer: the PPG12 analysis applies fiducial cuts $|z_r| < 60\,\mathrm{cm}$ on the reco vertex and $|z_h| < 60\,\mathrm{cm}$ on the primary truth vertex (Section 2), and inside those cuts the per-bin closure is already at the 1–2 % level. The stuck edge bin at $|z_r| \simeq 83\,\mathrm{cm}$ is well outside the analysis fiducial, so the iteration is good enough for the analysis use case, even though it is not formally converged on the full closure-window metric.

**(iii) Downstream hookup is wired in but off by default.** Both `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` now accept two new analysis fields, `truth_vertex_reweight_on` and `truth_vertex_reweight_file`; when `on=1`, the code opens `reweight.root`, reads `h_w_iterative`, and multiplies the per-event weight by $w(z_h)$ for single-interaction MC and $w(z_h)\,w(z_{\mathrm{mb}})$ for double-interaction MC. The secondary truth vertex branch `vertexz_truth_mb` (i.e. $z_{\mathrm{mb}}$) is picked up through an optional-pointer guard, so single-MC trees without it fall through to the single-vertex path. The two reco-level and truth-level reweights are mutually exclusive and the config loader enforces this by forcing `vertex_reweight_on=0` when `truth_vertex_reweight_on=1`. The showershape configs `config_showershape_{1p5rad,0rad}.yaml` are pre-wired to the matching `output/{1p5mrad,0mrad}/reweight.root`; the nominal BDT configs carry the same fields disabled, so flipping the main analysis over to the truth-vertex reweight is a one-line config edit once the jet-sample extension is done.

**(iv) Implementation can be sped up.** Each iteration currently rebuilds the MC reco histogram from scratch ($\mathcal{O}(N_{\mathrm{events}})$ per pass). Precomputing $z^2$ and dropping the per-trial mask rebuild would give a $\sim 10\times$ speedup with no change in result. Not a blocker; nice-to-have for the all-samples extension.

## 7 Summary

$f(z_r)$ cannot close the individual truth-vertex distributions ($z_h$ and $z_{\mathrm{mb}}$) of double-interaction MC, which biases the primary truth fiducial cut $|z_h| < 60\,\mathrm{cm}$ used in the efficiency denominator by $+1.4\,\%$ at 1.5 mrad (Section 2). An iterative data-driven reweight $w(z)$, applied per truth vertex with the factorization $w(z_h)\,w(z_{\mathrm{mb}})$ for double-interaction events, closes this gap without any Gaussian assumption. Applied to the photon10 mix, the method converges cleanly at 0 mrad ($\max|r-1| < 1\,\%$ in 8 iterations) and reduces in-window $\chi^2$ by a factor $\sim$3100 at 1.5 mrad, closing the analysis-relevant bulk $|z_r| < 30\,\mathrm{cm}$ to 1–2 % per bin and reaching $\sim$11 % at the closure-window

edge. The recovered $w(z)$ at 1.5 mrad shows non-Gaussian features (a strong central boost and a mid-$z$ shoulder) that no closed-form Gaussian ansatz could represent. Per-period `reweight.root` outputs are ready in `efficiencytool/truth_vertex_reweight/output/` and the C++ hookup into `ShowerShapeCheck.C` and `RecoEffCalculator_TTreeReader.C` is in place (enabled on the showershape period configs, wired but off on the nominal BDT configs). The remaining work is the jet-sample extension of the standalone fit tool.

# A Baseline: events with and without a reconstructed vertex

Before building any truth-based reweight, we checked that dropping events whose reco MBD vertex is the $-9999$ sentinel does not itself bias the MC. The concern is that an MC event which loses its reco vertex $z_r$ may be physically unusual (e.g. with a shifted truth vertex $z_h$), so dropping it would skew the efficiency in a $z_h$-dependent way.

| Sample | $N_{\text{total}}$ | $N_{\text{reco}}$ | $N_{\text{no reco}}$ | fraction no reco |
|---|---|---|---|---|
| photon10 | 9 998 562 | 7 716 585 | 2 281 977 | **22.82 %** |
| photon10_double | 9 997 075 | 9 491 223 | 505 852 | **5.06 %** |
| jet12 | 10 000 000 | 7 689 530 | 2 310 470 | **23.10 %** |
| jet12_double | 9 997 000 | 9 485 384 | 511 616 | **5.12 %** |

Table 2: Fraction of events without an MBD reco vertex. The double-interaction productions have $\sim 4.5\times$ higher MBD vertex-finding efficiency than their single-interaction counterparts (two concurrent collisions give the MBD twice the chance of producing a vertex).

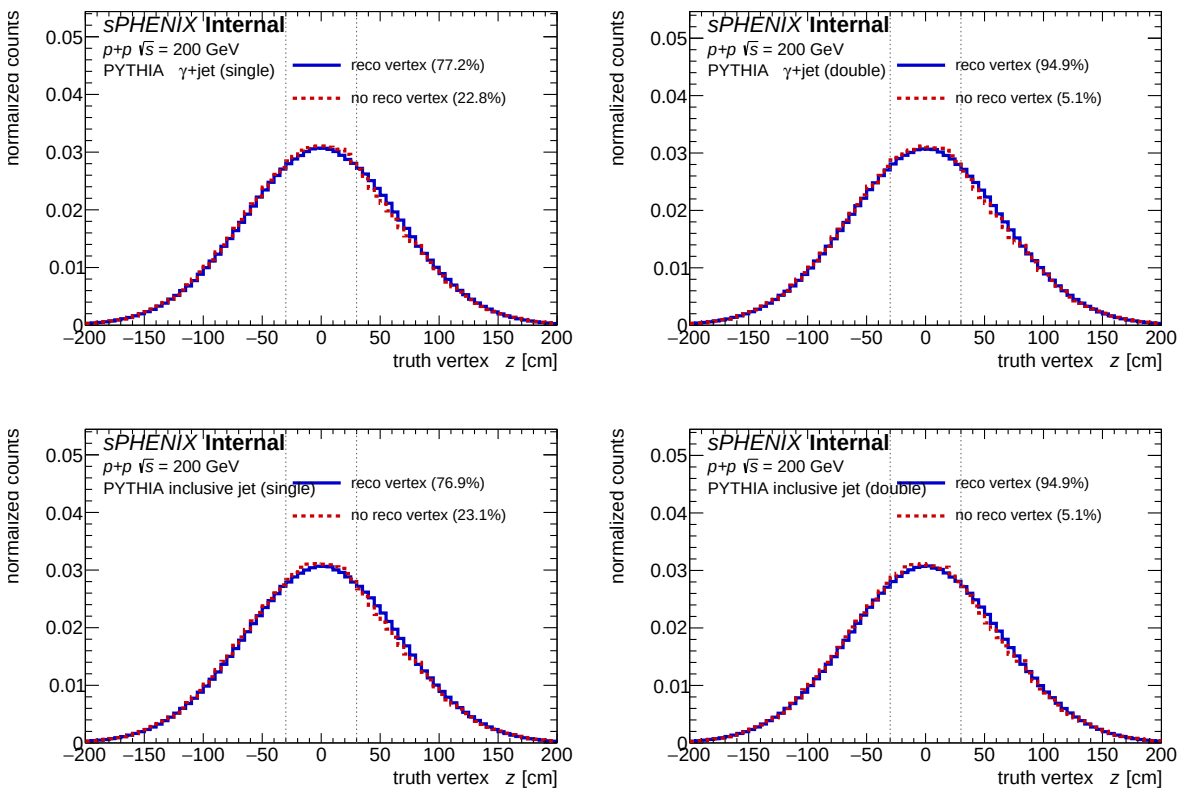


Figure 7: Primary truth vertex $z_h$ distribution for events with (blue) and without (red dashed) a reconstructed MBD vertex $z_r$, area-normalized for shape comparison. Columns: single / double interaction. Rows: photon10 / jet12. Dotted vertical lines: $|z_h| = 30$ cm reference.

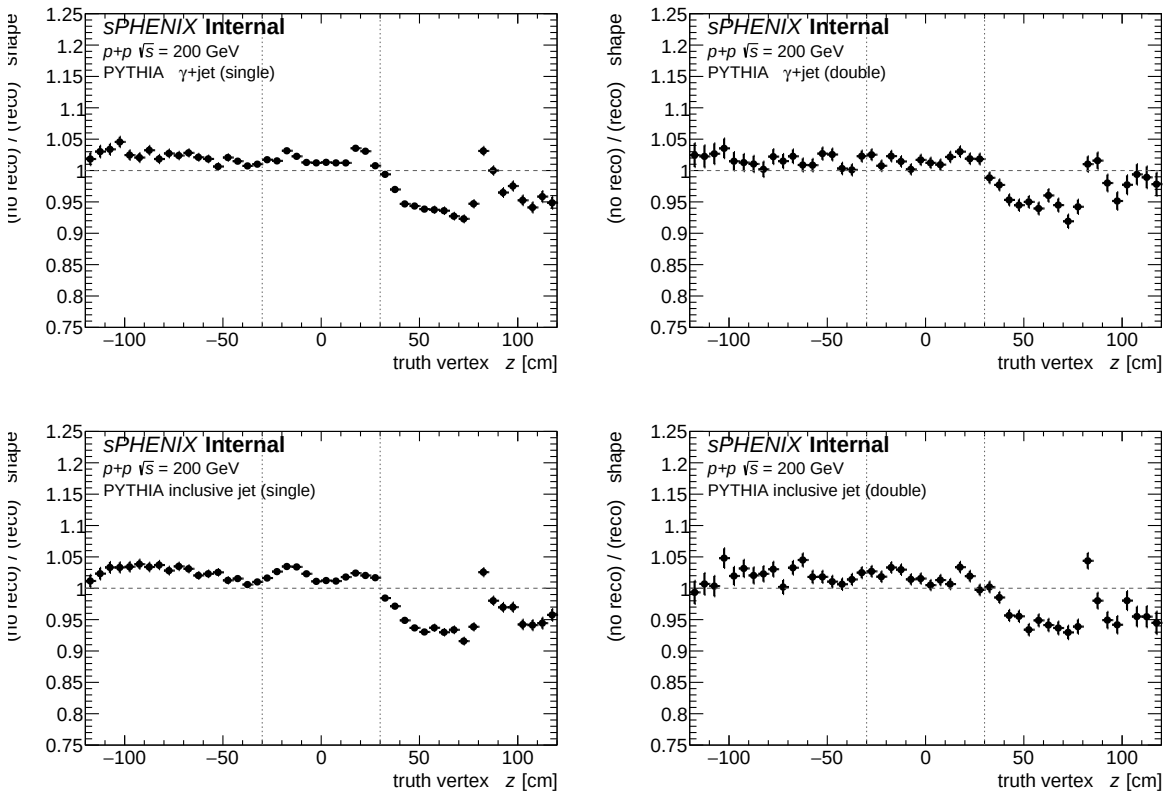


Figure 8: Ratio (no-reco-vertex shape) / (reco-vertex shape) as a function of primary truth vertex $z_h$, zoomed to $[0.75, 1.25]$. All samples show a $\sim\pm5\,\%$ north/south asymmetry inside $|z_h| < 30\,\mathrm{cm}$, consistent with a small MBD geometric bias.

Inside $|z_h| < 30\,\mathrm{cm}$ the two shapes agree to within $\pm5\,\%$ with a small $z_h$-asymmetry that is the same in all four samples and therefore a detector-level effect (MBD vertex finder slightly more efficient at positive $z$). Dropping the no-reco-vertex events is thus *approximately* harmless — both the efficiency numerator and denominator are scaled by the same factor, and the residual $z$ dependence of the drop rate is small. However, dropping $\sim 22\,\%$ of the single-interaction MC sample is still a lot of statistics, and it makes the efficiency measurement implicitly conditional on a valid reco vertex being found. That was the original motivation for exploring a truth-level reweight in the first place.

# B Discarded methods

Before arriving at the iterative method described in Section 3, we explored three truth-only reweight constructions that did not work:

- $w_1$**: direct ratio** of the data reco-vertex shape to the MC truth-vertex shape, $w_1(z_h) = D(z_r)/T(z_h)$, with the common binning identifying $z_r$ bins with $z_h$ bins.
- $w_2$**: conditional projection**, $w_2(z_h) = \mathbb{E}[f(z_r) \,|\, z_h]$, the expectation of the production reco weight conditional on the primary truth vertex.
- $w_{\mathrm{iter,DA}}$**: D'Agostini iterative unfolding** using the MC response matrix $R_{ij} = P(z_r^i|z_h^j)$.

All three hit a geometric lower bound on the double-MC sample at 1.5 mrad: the conditional reco kernel $P(z_r|z_h)$ (for single-interaction MC) has RMS $\sim 34\,\mathrm{cm}$, wider than the $\sim 22\,\mathrm{cm}$ target data peak (Section C), so *any* truth-only reweight is bounded from below by the kernel width and cannot reach the target.

We also drafted a closed-form **2D Gaussian reweight** $w(z_h, z_{\mathrm{mb}}) \propto \exp[-\frac{1}{2}(z_h^2 + z_{\mathrm{mb}}^2)\,\Delta]$ with $\Delta = 1/\sigma_{\mathrm{lum}}^2 - 1/\sigma_{\mathrm{MC}}^2$. This closed form does work in principle — it acts on both the symmetric and antisymmetric coordinates and so can narrow all three vertex axes simultaneously — but it assumes a single Gaussian shape for the data luminous region. At 1.5 mrad the data has a $\pm$-asymmetric mid-tail bump that a symmetric Gaussian cannot reproduce, so the closed form ended up as a first-draft proposal that was superseded by the iterative method of Section 3, which discovers $w(z)$ from data directly without any Gaussian assumption.

# C Double-interaction reco-truth kernel

The geometric lower bound on truth-only reweights quoted in Section B can be read off the widths directly:

- 1.5 mrad data reco width: $\sigma \simeq 22.3\,\mathrm{cm}$
- jet12 single-interaction MC truth width ($z_h$): $\sigma \simeq 57.6\,\mathrm{cm}$
- jet12_double MC truth width ($z_h$, primary): $\sigma \simeq 62.7\,\mathrm{cm}$
- jet12_double MC reco width ($z_r$): $\sigma \simeq 48.6 \approx 62.7/\sqrt{2}$ (averaging)
- Conditional kernel $P(z_r|z_h{=}0)$ RMS (with $z_{\mathrm{mb}}$ marginalized): $\sim 34\,\mathrm{cm}$

Figure 9 shows the kernel slice directly, compared to the 1.5 mrad data target. Even if the primary truth vertex is forced to $z_h = 0$, the reco distribution from a double-interaction event cannot be narrower than the $\sim 34\,\mathrm{cm}$ kernel width — larger than the 22 cm target. This is the structural reason the truth-only methods fail on double-interaction MC, and the structural reason the iterative method needs the $w(z_h)\,w(z_{\mathrm{mb}})$ factorization rather than a $w$-on-$z_r$ construction.

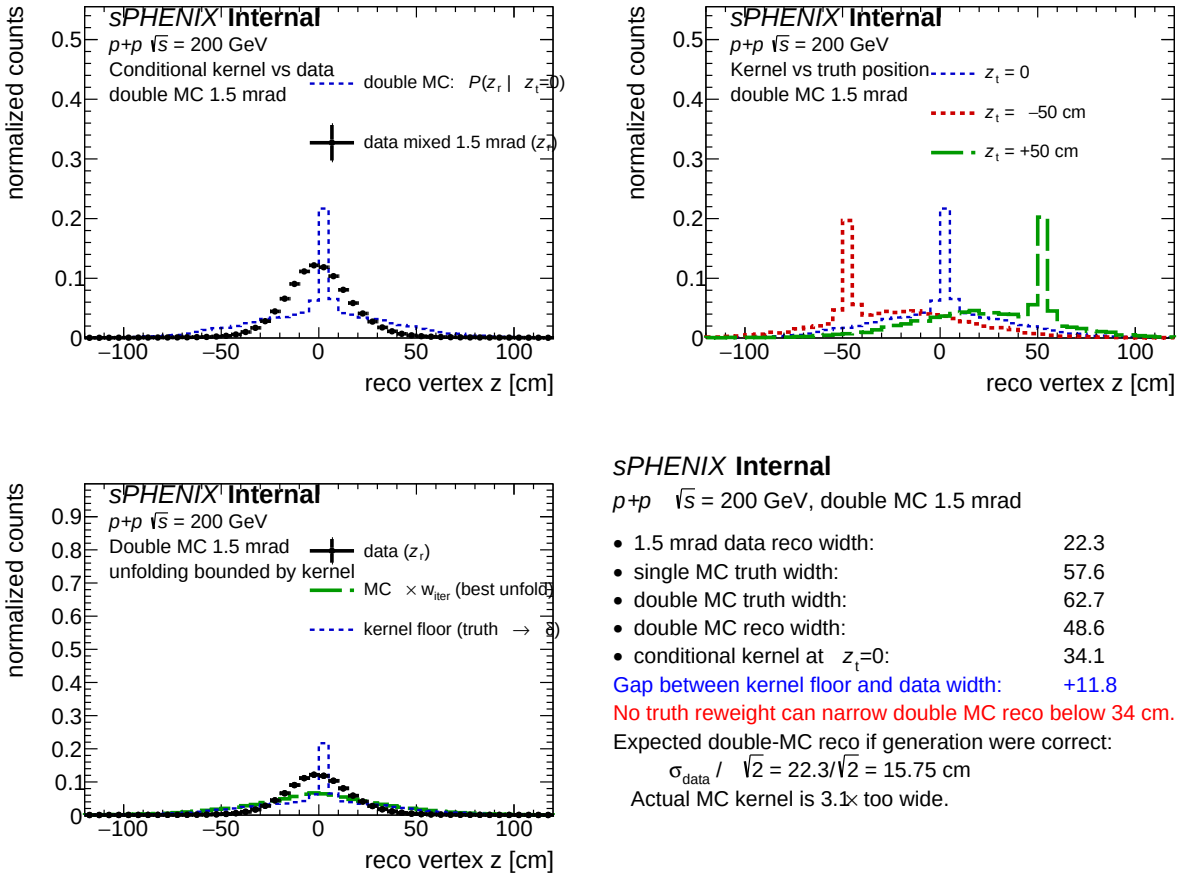


Figure 9: Conditional reco kernel $P(z_r|z_h)$ in jet12_double MC at 1.5 mrad (with $z_{\mathrm{mb}}$ marginalized), with the 1.5 mrad data reco $z_r$ shape overlaid. The kernel's $\sim$34 cm RMS exceeds the $\sim$22 cm data peak, which is the geometric lower bound on truth-only reweighting.

# D Files and code

- `anatreemaker/source/CaloAna24.{h,cc}` — writes the `vertexz_truth` ($z_h$, primary truth vertex) and `vertexz_truth_second` ($z_{\mathrm{mb}}$, secondary truth vertex; read by C++ as `vertexz_truth_mb`) branches that the iterative method reads for single- and double-interaction MC respectively.
- `efficiencytool/truth_vertex_reweight/` — the standalone Python tool for the iterative method:
  - `fit_core.py` — cached sample loader, the `mixed_mc_histogram` forward model (Eq. 2), Poisson $\chi^2$, and helpers for fitting $\sigma_{\mathrm{gen}}$ from MC truth.
  - `fit_truth_vertex_reweight.py` — the driver (stage iterative = Section 3), producing `reweight.root` and `fit_result.yaml`.
  - `plot_closure.py`, `plot_comparisons.py` — closure and cross-period diagnostic plots.
  - `config.yaml` — sample paths, run ranges, fit parameters.
- `efficiencytool/TruthVertexRecoCheck.C` — Appendix A supporting macro.
- Outputs: `efficiencytool/truth_vertex_reweight/output/{1p5mrad,0mrad}/reweight.root`, `fit_result.yaml`, and the cached event arrays in `output/cache/`.